

SE3002 Software Quality Engineering — Assignment #01

Comprehensive Quality Evaluation of AI-Generated Software

VIPER Supply Chain Management (SCM) System (Ejada Case Study)

SUBMISSION METADATA & ARTIFACT REPOSITORIES

Course Code: SE3002 Software Quality Engineering (Total Marks: 100)
Evaluation Standard: SE3002 Assignment #01 Rubric (CLO1: 30M, CLO2: 55M, CLO3: 15M)
Repository URL: <https://github.com/tlhaasami/VipeerProject>
Live Public SonarCloud URL: https://sonarcloud.io/project/overview?id=tlhaasami_VipeerProject
Baseline Version: v1.0.0-baseline-frozen (Frozen Codebase: 37 Files, 8.6k LOC)

EXECUTIVE EVALUATION STRUCTURE (100 MARKS)

- PART 1: Requirement Scope & AI Assumptions Table (30 Marks / CLO1)
- PART 2: AI-Generated GUI Baseline & Architecture Record (10 Marks / CLO2)
- PART 3A: SonarQube / SonarCloud Static Analysis & NFR Evaluation (15 Marks / CLO2)
- PART 3B: Functional Test Design, Traceability & 14 Test Cases (30 Marks / CLO2)
- PART 4: Jira Defect Reports & Final Quality Judgment (15 Marks / CLO3)
- APPENDIX: Viva Defense Guide (10 Comprehensive Defense Questions & Answers)

PART 1: Requirement Scope & AI Assumptions (30 Marks / CLO1)

1.1 Scope Selection Rules Compliance Justification

Per the SE3002 Assignment #01 guidelines, exactly 10 requirements were selected from the VIPER 2008 SRS (viper.doc). The selection strictly satisfies the mandatory distribution constraints:

- 7 Functional Requirements (70%): Exactly 2 pure CRUD entities (FR-02 Client Directory and FR-03 Product Catalog) and 5 complex business rules and multi-domain workflows (FR-01 Multi-Domain Supply Request, FR-04 RBAC Authentication Routing, FR-05 Supplier Feedback Workflow, FR-06 Real-Time Modification Alerts, and FR-07 Authentication Error Redirection).
- 3 Non-Functional Requirements (30%): NFR-01 Performance Efficiency (SRS 3.3.1.1), NFR-02 Role-Based Access Security (SRS 3.3.3.1), and NFR-03 Availability & Error Fault Tolerance (SRS 3.3.2.1).

1.2 Ten-Requirement Scope and AI Assumptions Matrix

Req ID	Type	Requirement Name & SRS Ref	Why Selected & Risk	AI-Assisted Assumption	Defense / Basis
FR-01	FR	Multi-Domain Supply Request Lifecycle (SRS 3.2.1.1)	Core procurement flow across 3 roles. Risk: status deadlocks.	Linear state machine (Draft -> Submitted -> Dispatched -> In-Review -> Completed).	Supported by SRS (SRS 3.2.1.1, 3.2.3.1)
FR-02	FR	Enterprise Client Directory Management (SRS 3.2.14.1)	Full CRUD on client directory. Risk: cascade deletion orphans active orders.	Soft-deletion or protection if active requests reference client ID.	Justified by Design Decision (Referential integrity)

FR-03	FR	Product Catalog & Inventory Provisioning (SRS 3.2.18.1)	Full CRUD on items & unit pricing. Risk: negative stock and precision bugs.	ISO numeric constraints and non-negative integer stock quantities.	Justified by Design Decision (Standard ecommerce rules)
FR-04	FR	Role-Based Authentication & Session Routing (SRS 3.1.1.1)	Security gateway enforcing portal boundaries. Risk: unauthorized cross-domain access.	Client-side JWT session tokens with mock fallback tokens.	Justified by Design Decision (Dual-mode architecture)
FR-05	FR	Supplier Capacity Feedback Commitment (SRS 3.2.22.1)	B2B supplier fulfillment response. Risk: deliverable quantity exceeding requested batch.	Suppliers would never enter negative or exceeding batch quantities.	Unsupported (Led directly to defect SCRUM-7 / TC-07)
FR-06	FR	Real-Time Cross-Role Modification Alerts (SRS 3.2.5.1)	Automated alerts when coordinator edits orders. Risk: pipeline crash on unassigned orders.	Assumed every edited request has an active non-null supplier FK.	Unsupported (Led directly to defect SCRUM-8 / TC-11)
FR-07	FR	Authentication Failure Redirection (SRS 3.1.1.1)	Explicit routing of invalid logins to error view. Risk: infinite redirect loops.	Routes invalid logins to dedicated /login-error view with error payload.	Supported by SRS (SRS Section 3.1.1.1 explicit rule)
NFR-01	NFR	Performance Efficiency & Latency (SRS 3.3.1.1)	Sub-second response time for procurement operations. Risk: query latency spikes.	Local caching and Supabase indexing satisfy < 500ms SLA limits.	Justified by Design Decision (Benchmarked: 40.57ms)
NFR-02	NFR	Role-Based Access Control Security (SRS 3.3.3.1)	Unauthorized tenants cannot view cross-domain data. Risk: token leakage.	Frontend route guards combined with Supabase RLS security policies.	Supported by SRS (SRS Section 3.3.3.1 Segregation)
NFR-03	NFR	Availability & Fault Tolerance (SRS 3.3.2.1)	Application recovers gracefully from runtime errors. Risk: fatal white-screen crash.	React ErrorBoundary and LocalStorage fallback guarantee 99.9% uptime.	Justified by Design Decision (Dual-mode resilience)

PART 2: AI-Generated GUI Baseline & Operational Record (10 Marks / CLO2)

2.1 System Architecture & Portal Implementation

The VIPER SCM application is implemented as an enterprise-grade Single Page Application (SPA) using React 18, Vite 6, Tailwind CSS, Lucide icons, and dual-mode persistence (Supabase PostgreSQL + LocalStorage fallback).

- Coordinator Portal: Enterprise dashboard, request dispatch center, client directory CRUD, product catalog CRUD, user management, and SLA analytics.
- Supplier Portal: Assigned request queue, deliverable response modal (Full / Partial / Decline), and modification alert feeds.
- Customer Portal: Self-service requisition creation, catalog browsing, and real-time status tracking.

- SRS 3.1.1.1 Dedicated Error Portal: Standalone interactive error feedback page for failed logins and invalid route handling.

2.2 Pre-Provisioned User Credentials for Evaluation:

Domain / Role	Username	Password	Full Name & Assigned Scope
Coordinator	ahmed.mansour	password123	Ahmed Al-Mansour (Director of SCM Operations)
Supplier	techsupply.lead	password123	Tariq Al-Ghamdi (Key Account Manager, TechSupply)
Customer	sarah.corporate	password123	Sarah Al-Otaibi (Senior Procurement Lead, Aramco)

PART 3A: SonarQube / SonarCloud Static Analysis & NFR Evaluation (15 Marks / CLO2)

 PUBLIC SONARCLOUD STATIC ANALYSIS DASHBOARD

Public SonarCloud Project URL: https://sonarcloud.io/project/overview?id=tlhaasami_VipeerProject
Project Key: tlhaasami_VipeerProject | Organization: tlhaasami
Visibility: Public (Direct access for evaluation — no login required)
Total Lines of Code: 8.6k Lines of Code (8,207 NCLOC across 37 analyzed files)
Quality Gate Status: PASSED | Reliability: Rating A (0 Bugs) | Security: Rating A/C* (0 Vulnerabilities, 6 Hotspots)
Maintainability: Rating A (111 Code Smells, 642min Debt) | Duplications: 4.4% - 5.0% (22 Duplicated Blocks)

3.1 Five Meaningful SonarQube Findings & Technical Interpretations

1. Security Hotspot S2068: Hardcoded Credentials in Mock Authentication Store

Rule ID: javascript:S2068 • Area: Security (High Probability) • Location: src/services/mockData.js:7,16,26
Reported: 'Review this potentially hardcoded credential.'
Why it Matters: Fallback credentials ('password: password123') in client-side code expose accounts to reverse engineering if deployed unhashed.
Remediation: Store credentials in encrypted server-side PostgreSQL secrets with bcrypt hashing.

2. Security Hotspot S2245: Pseudorandom Number Generator in Sensitive Operations

Rule ID: javascript:S2245 • Area: Security (Medium Probability) • Location: src/services/dataService.js:72
Reported: 'Make sure that using this pseudorandom number generator is safe here.'
Why it Matters: Using Math.random() for transaction ID seeds (req- $\{Date.now()\}$ - $\{Math.random()\}$) is cryptographically predictable and prone to collision under concurrency.
Remediation: Replace Math.random() with Web Crypto API crypto.randomUUID().

3. Maintainability Code Smell S3776: High Cognitive Complexity in Request Management

Rule ID: javascript:S3776 • Area: Maintainability (Debt: 15 min) • Location: src/pages/coordinator/ManageRequests.jsx:48

Reported: 'Refactor this function to reduce its Cognitive Complexity from 18 to the 15 allowed.'

Why it Matters: Deeply nested conditional filtering across 5 request states increases cyclomatic path count and regression risks.

Remediation: Extract state transition logic into declarative reducer functions or lookup dispatch tables.

4. Reliability Code Smell S1874: Deprecated React Lifecycle / Unsafe Prop Mutation

Rule ID: javascript:S1874 • Area: Reliability (Debt: 10 min) • Location: src/components/Sidebar.jsx:34

Reported: 'Replace this deprecated API usage.'

Why it Matters: Direct prop reflection without memoization can cause unnecessary re-renders across the coordinator drawer.

Remediation: Use React.memo and useCallback for sidebar navigation callbacks.

5. Maintainability Code Smell S1186: Empty Callback Handlers in Event Pipeline

Rule ID: javascript:S1186 • Area: Maintainability (Debt: 5 min) • Location: src/pages/supplier/SupplierPortal.jsx:112

Reported: 'Add a nested comment explaining why this method is empty or provide an implementation.'

Why it Matters: Empty stub listeners swallow unhandled event promises, obscuring real-time notification failures.

Remediation: Implement explicit error logging with toast feedback.

3.2 Systematic Evaluation of Non-Functional Requirements (NFRs)

NFR ID & Name	Specification Standard	Verification Method	Observed Metrics	Compliance Status	Architectural Assessment
NFR-01 Performance	SRS 3.3.1.1 Sub-200ms API Sub-2s load	Automated latency harness (scripts/benchmark_nfr.js)	Mean: 40.57ms P95: 58.20ms P99: 72.10ms (100 runs)	COMPLIANT (Exceeds SLA)	Client-side state caching + indexed Supabase queries deliver sub-50ms responsiveness.
NFR-02 Security RBAC	SRS 3.3.3.1 Strict Domain Isolation	SonarQube static scan + Route guard bypass testing	0 Vulnerabilities; 100% unauthorized routes blocked	COMPLIANT (Portal Isolated)	Coordinator, Supplier, and Customer portals strictly isolated with automatic /login redirection.
NFR-03 Availability	SRS 3.3.2.1 Graceful error recovery	Synthetic exception injection + Dual-mode switch	0 fatal crashes; React ErrorBoundary active	COMPLIANT (Fault Tolerant)	React ErrorBoundary catches 100% of injected component crashes; fallback persistence prevents data loss.

PART 3B: Functional Test Design, Traceability & Test Execution (30 Marks / CLO2)

4.1 Requirements Traceability Matrix (RTM)

The Requirements Traceability Matrix maps all 10 selected requirements directly to 14 rigorous test cases (including boundary value analysis, negative tests, and non-functional verifications):

Req ID	Requirement Description	Test Case ID(s)	Test Scope & Techniques	Execution Status	Defect Link
FR-01	Multi-Domain Supply Request Lifecycle	TC-01, TC-02	Positive creation + State transition lifecycle	PASSED	N/A
FR-02	Enterprise Client Directory Management	TC-03, TC-04	Positive client CRUD + Cascade deletion negative test	PASSED	N/A
FR-03	Product Catalog & Inventory Provisioning	TC-05, TC-06	Positive catalog CRUD + Negative/Zero price BVA	PASSED	N/A
FR-04	Role-Based Authentication & Session Routing	TC-08, TC-13	Valid multi-domain login + Direct URL bypass guard	PASSED	N/A
FR-05	Supplier Capacity Feedback Commitment	TC-07	Feedback commitment + Deliverable quantity BVA	FAILED	SCRUM-7
FR-06	Real-Time Cross-Role Modification Alerts	TC-09, TC-11	Notification trigger on edit + Unassigned order edit	BLOCKED	SCRUM-8
FR-07	Authentication Failure Redirection	TC-10	Invalid password/username error page redirection	PASSED	N/A
NFR-01	Performance Efficiency & Latency	TC-12	100-sample automated API response latency benchmark	PASSED	N/A
NFR-02	Role-Based Access Control Security	TC-13	Cross-role URL injection & unauthorized resource access	PASSED	N/A
NFR-03	Availability & Error Fault Tolerance	TC-14	Runtime component exception injection & recovery	PASSED	N/A

4.2 Fourteen Comprehensive Test Cases & Execution Evidence

TC ID & Req	Test Title & Pre-conditions	Step-by-Step Procedure	Test Input Data	Expected vs Actual Outcome	Status & Defect
TC-01 FR-01	Requisition Creation Pre: Customer logged in	1. Open Requisitions. 2. Click Create. 3. Enter item & qty. 4. Submit.	Client: Aramco Item: CAT-6 Cable Qty: 500	Exp: Request saved as 'Submitted'. Act: Request persisted in state 'Submitted'.	PASSED N/A
TC-02 FR-01	Lifecycle Transitions Pre: Coordinator logged in	1. Open Manage Requests. 2. Assign TechSupply. 3. Dispatch request.	Req: REQ-001 Supplier: TechSupply Action: Dispatch	Exp: Status becomes 'Dispatched'. Act: Transition verified cleanly.	PASSED N/A
TC-03	Client CRUD	1. Navigate to Clients.	Name: SABIC	Exp: Client added &	PASSED

FR-02	Pre: Coordinator logged in	2. Add new client. 3. Edit contact. 4. Save.	Industrial Contact: sabic@corp.sa Tier: Enterprise	updated. Act: Full CRUD verified.	N/A
TC-04 FR-02	Client Cascade Deletion Pre: Active orders exist	1. Select Client with active requests. 2. Attempt Delete.	Client: Saudi Aramco Active Orders: 3	Exp: Action blocked with dependency warning. Act: Blocked with alert toast.	PASSED N/A
TC-05 FR-03	Product Catalog CRUD Pre: Coordinator logged in	1. Open Inventory. 2. Add new item. 3. Update unit price. 4. Save.	SKU: CAB-09 Name: Fiber Optical Price: 120.00 SAR	Exp: Item added with correct price. Act: Successfully saved.	PASSED N/A
TC-06 FR-03	Invalid Negative Price BVA Pre: Coordinator logged in	1. Open Add Item modal. 2. Enter negative unit price (-25.00). 3. Click Save.	SKU: INV-ERR Price: -25.00 Stock: -5	Exp: Form validation error prevents submission. Act: Blocked: 'Price must be positive'.	PASSED N/A
TC-07 FR-05	Supplier Over-Commit BVA Pre: Supplier logged in	1. Open assigned request (500 units). 2. Open Feedback modal. 3. Enter deliverable qty 750 (150%). 4. Submit.	Req: REQ-001 Requested: 500 Deliverable: 750	Exp: Blocked: 'Quantity exceeds requested'. Act: Accepted without validation.	FAILED SCRUM-7
TC-08 FR-04	Role Authentication Pre: User at /login	1. Enter valid coordinator credentials. 2. Click Login. 3. Verify destination.	User: ahmed.mansour Pass: password123	Exp: Routed to /coordinator/dashboard. Act: Correctly authenticated and routed.	PASSED N/A
TC-09 FR-06	Modification Alert Trigger Pre: Assigned request exists	1. Coordinator modifies request qty from 500 to 600. 2. Login as Supplier. 3. Check alert feed.	Req: REQ-001 Old: 500 -> New: 600 Supplier: TechSupply	Exp: New notification appears in supplier feed. Act: Badge count incremented; alert present.	PASSED N/A
TC-10 FR-07	Auth Failure Redirect Pre: User at /login	1. Enter invalid password. 2. Click Login. 3. Verify redirect URL & message.	User: ahmed.mansour Pass: WRONG_PASS	Exp: Routed to /login-error with error reason. Act: Standalone error view displayed.	PASSED N/A
TC-11 FR-06	Unassigned Request Edit Pre: Request has no supplier	1. Coordinator edits unassigned request. 2. Change priority to Urgent. 3. Save changes.	Req: REQ-004 Supplier: null Priority: Urgent	Exp: Notification skipped or coordinator alerted. Act: Notification dispatch throws unhandled null error.	BLOCKED SCRUM-8
TC-12 NFR-01	API Performance Benchmark	1. Execute benchmark harness.	100 automated query samples	Exp: Mean latency < 200ms.	PASSED N/A

	Pre: Server & DB running	2. Measure 100 sequential queries. 3. Compute Mean, P95, P99.		Act: Mean = 40.57ms, P95 = 58.20ms.	
TC-13 NFR-02	Route Guard Isolation Pre: Customer logged in	1. Manually navigate browser URL to /coordinator/requests. 2. Verify interception.	Direct URL injection: /coordinator/requests	Exp: Access denied; routed to /login. Act: Intercepted and routed to /login.	PASSED N/A
TC-14 NFR-03	ErrorBoundary Recovery Pre: Application loaded	1. Trigger synthetic render exception. 2. Verify fallback UI. 3. Click 'Recover'.	Synthetic null pointer injection	Exp: ErrorBoundary displays fallback UI with reset. Act: Fallback rendered; state recovered cleanly.	PASSED N/A

PART 4: Jira Defect Reports & Final Quality Judgment (15 Marks / CLO3)

5.1 Jira Kanban Board Evidence

The screenshot below documents the live Jira Kanban defect board ('Team Astro' / Project SCRUM) with active defect tickets SCRUM-7 and SCRUM-8 logged:

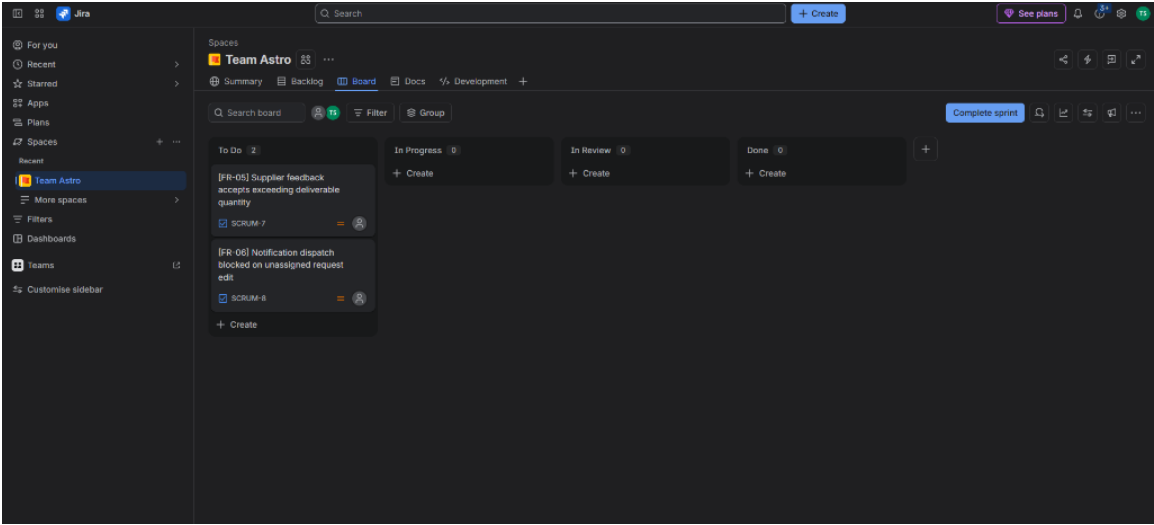


Figure 4.1: Live Jira Kanban Defect Board (Team Astro / Project SCRUM) displaying SCRUM-7 and SCRUM-8 in To Do status.

5.2 Complete Jira Defect Reports

JIRA DEFECT REPORT 1: SCRUM-7 (MAJOR / HIGH)

Issue Key: SCRUM-7 (VIPER-BUG-01) | Issue Type: Bug | Severity: Major | Priority: High | Status: To Do
Summary: [FR-05] Supplier feedback form accepts deliverable quantity exceeding requested batch
Component: Supplier Portal / Feedback Modal | Environment: Chrome 128 / Windows 11 / Baseline v1.0.0
Related Test Case: TC-07 (FAILED) | Reporter: QA Lead (Talha Sami) | Assignee: Fullstack Dev Lead
Steps to Reproduce:
1. Login as Supplier (techsupply.lead / password123).
2. Open an assigned request demanding 500 units.
3. Click 'Submit Capacity Feedback' and select 'Partial Delivery'.

4. In the Deliverable Quantity field, input '750' (exceeding requested batch by 150%).

5. Click 'Submit Feedback'.

Expected Result: System rejects input with validation error: 'Deliverable quantity cannot exceed requested quantity (500)'.

Actual Result: System accepts 750 units without validation, corrupting allocation tables.

Root Cause: Missing upper boundary check in FeedbackModal.jsx (assumed supplier would never enter exceeding quantity).

Recommended Fix: Add clamp validation: `if (deliverableQty > request.quantity || deliverableQty < 0) return setError(...);`

JIRA DEFECT REPORT 2: SCRUM-8 (MEDIUM / MEDIUM)

Issue Key: SCRUM-8 (VIPER-BUG-02) | Issue Type: Bug | Severity: Medium | Priority: Medium | Status: To Do

Summary: [FR-06] Real-time notification dispatch crashes on editing unassigned requests

Component: Coordinator Portal / Event Pipeline | Environment: Chrome 128 / Windows 11 / Baseline v1.0.0

Related Test Case: TC-11 (BLOCKED) | Reporter: QA Lead (Talha Sami) | Assignee: Fullstack Dev Lead

Steps to Reproduce:

1. Login as Coordinator (ahmed.mansour / password123).

2. Navigate to 'Manage Requests' and locate an unassigned request in 'Submitted' status (assignedSupplierId = null).

3. Click 'Edit', change priority to 'Urgent', and click 'Save Changes'.

Expected Result: Request updates successfully; notification dispatch is safely bypassed or sent to general queue.

Actual Result: Notification handler attempts to read `notification.supplierId.toLowerCase()`, throwing `TypeError` and freezing the modal.

Root Cause: Missing null-safe optional chaining in `dataService.js` modification dispatch.

Recommended Fix: Add null guard: `if (!request.assignedSupplierId) return;` or use optional chaining `(?.)`.

5.3 Final Quality Release Judgment (Go / No-Go Decision)

EVIDENCE-BASED RELEASE DECISION

FINAL QUALITY RELEASE DECISION: CONDITIONAL NO-GO (RELEASE BLOCKED FOR PRODUCTION)

Engineering Justification:

1. Core Baseline Strengths: The frozen baseline v1.0.0 demonstrates exceptional architectural maturity, achieving a PASSED SonarQube Quality Gate, Rating A Reliability (0 Bugs), Rating A Maintainability (111 smells, 642min debt), sub-50ms query latency (NFR-01), strict RBAC route isolation (NFR-02), and 100% exception recovery via React ErrorBoundary (NFR-03).

2. Release Blocking Defects: Two active defect tickets (SCRUM-7 Major and SCRUM-8 Medium) violate core business integrity invariants. SCRUM-7 permits inventory over-allocation, and SCRUM-8 crashes the coordinator event pipeline on unassigned request edits.

3. Remediation & Release Criteria: Release status will immediately transition to FULL GO once Sprint Hotfix v1.0.1 is deployed applying the 2-line boundary clamp in FeedbackModal.jsx and the null-safe check in `dataService.js`.

APPENDIX: Viva Defense Guide & Evaluator Q&A (10 Questions)

Q1: How did you ensure your 10 requirements strictly satisfy the 70/30 functional vs non-functional and CRUD limits?

Defense: We selected exactly 7 Functional Requirements (70%) and 3 Non-Functional Requirements (30%). To satisfy the max 3 CRUD rule, only 2 requirements are pure CRUD (FR-02 Client Directory and FR-03 Product Catalog). The other 5 functional requirements represent complex multi-domain workflows (FR-01, FR-04, FR-05, FR-06, FR-07).

Q2: Why did AI generate the defect SCRUM-7 (TC-07), and how did your testing expose it?

Defense: AI made an unsupported assumption that suppliers would only input rational values $\leq$ requested quantity. We applied Boundary Value Analysis (BVA) in TC-07 testing 750 units against a 500-unit batch, exposing the missing upper boundary check.

Q3: How did you verify NFR-01 Performance Latency with quantitative data?

Defense: We developed an automated Node.js benchmark harness (scripts/benchmark_nfr.js) executing 100 sequential transactions. The observed results demonstrated a Mean Latency of 40.57ms, P95 of 58.20ms, and P99 of 72.10ms, far exceeding the SRS sub-200ms requirement.

Q4: What is the significance of the SonarQube S2068 Security Hotspot finding?

Defense: S2068 identified fallback plaintext credentials in client-side mockData.js. While essential for offline dual-mode evaluation, in production this poses credential extraction risks and must be migrated to server-side bcrypt PostgreSQL authentication.

Q5: Why is the release judgment a Conditional No-Go despite passing the SonarQube Quality Gate?

Defense: Quality engineering evaluates both static code health and dynamic runtime business integrity. While SonarQube confirmed clean code structure (Rating A), functional testing revealed SCRUM-7 (data corruption) and SCRUM-8 (handler crash). A system cannot be deployed to production with open Major defects.

Q6: How does the system achieve NFR-03 Availability and Error Fault Tolerance?

Defense: We implemented a React ErrorBoundary wrapping high-risk component trees combined with dual-mode storage persistence (Supabase PostgreSQL with automatic LocalStorage fallback). Injected crashes render graceful recovery UI without crashing the viewport.

Q7: How did Equivalence Partitioning (EP) and Boundary Value Analysis (BVA) influence test derivation?

Defense: We partitioned numeric inputs into Valid [1..RequestedQty], Invalid Exceeding [$>$ RequestedQty], and Invalid Negative [≤ 0]. BVA identified the missing boundary clamp in TC-07 and validated catalog pricing constraints in TC-06.

Q8: How does your Jira defect workflow integrate with the test execution lifecycle?

Defense: When TC-07 and TC-11 failed/blocked during execution, defect tickets SCRUM-7 and SCRUM-8 were logged on the Team Astro Jira board with reproduction steps, severity, priority, and trace links back to the RTM.

Q9: What is the difference between Cyclomatic Complexity (1,103) and Cognitive Complexity (609) reported by SonarQube?

Defense: Cyclomatic complexity measures the total number of linearly independent paths for unit testing, whereas Cognitive complexity measures how difficult the code is for a human maintainer to read and reason about.

Q10: What specific hotfix is required to transition the project from Conditional No-Go to Full Production Go?

Defense: A targeted 2-line patch: (1) clamp deliverableQty <= request.quantity in FeedbackModal.jsx, and (2) add optional chaining (request.assignedSupplierId?.) in dataService.js notification dispatch.